

Name: AMRITANSU SINGH AMRITANSU SINGH | Reg. No: RA2511003011028

## Unit – 1 : Session – 2 : SLO – 1 | Language Theory Concepts

### ■ Activity (Solutions)

- 1) Type checking verifies that operations are applied to compatible types. STATIC typing (Java): types are checked at compile time — `int x = 5; x = "hi";` fails to compile. DYNAMIC typing (e.g., Python): types are checked at run time — `x = 5; x = "hi"` is legal. Java demo: `String s = "abc"; int n = s.length();` // OK (`length()` returns `int`); `int n2 = s;` // compile-time type error.
- 2) Variable scope demo: `public class Scope { static int g = 10; // class scope public static void main(String[] a){ int x = 5; // method scope if (x > 0) { int y = 2; // block scope System.out.println(x + y + g); } /* y not visible here */ }`
- 3) Binding time vs run time: Binding time is when an association (name→type, variable→value, method→body) is made — e.g., '`int x`' binds `x` to type `int` at compile time; static methods bind at compile time. Run time is when the program executes — dynamic (polymorphic) method calls bind to the actual method body at run time. Example: `Animal a = new Dog(); a.sound();` — the reference type binds at compile time, but which `sound()` runs is decided at run time.